D.M. MEDICAL ONCOLOGY — PAPER IV

RECENT ADVANCES · EXAM ANSWER NOTES

Practice changing advances in early stage NSCLC.

January 2024 | DM 0124 | Q2 (DEEP Essay / Elaborate (FOCUS 2022–2026)) | **15 marks** | Target ~16 pages

*Includes: mechanisms · doses/regimens · landmark clinical trials · hallmarks · **NCCN 2026 / 2027 updates** · flow charts · exam pearls*

0. How to use this DEEP answer

Sections below are **high-yield, trial-dense content** written for a 15-mark Paper IV elaboration. After the deep core, additional revision modules (algorithms, supportive care, flash cards) extend the note to full booklet length.

EXAM PEARLS / HALLMARKS

- In the exam hall, prioritise the DEEP trial tables + NCCN 2026 boxes first.
- Draw at least one flowchart from the deep sections.

1. Introduction & exam framing

Early-stage NSCLC (AJCC resectable I–III and unresectable stage III) has been transformed by adjuvant EGFR TKI (ADAURA), consolidation PD-L1 after CRT (PACIFIC), EGFR TKI after CRT (LAURA), and perioperative immunotherapy (CheckMate-816, KEYNOTE-671, AEGEAN). Molecular testing and AJCC 9th staging (NCCN NSCLC v9.2026) are now integral even before surgery or CRT.

Answer plan: staging/molecular prerequisites → stage-by-stage advances with trial numbers → perioperative IO comparison → adjuvant IO → unresectable III pathways → doses → NCCN 2026 → pearls.

- Always bifurcate EGFR/ALK driver+ vs driver– pathways.
- Distinguish adjuvant (post-op) vs neoadjuvant vs perioperative (pre+post) designs.
- Quote EFS/DFS/OS endpoints carefully — examiners dock loose language.

EXAM PEARLS / HALLMARKS

- Opening line: 'Early NSCLC is no longer surgery ± chemo alone — biomarker-directed adjuvant/perioperative therapy is standard.'

2. Staging update — AJCC 9th & work-up

NCCN NSCLC v9.2026 adopts AJCC 9th edition staging. Practical work-up: contrast CT, PET-CT for staging, brain MRI for stage II–III (and selected IB), invasive mediastinal staging (EBUS/mediastinoscopy) when it changes therapy, and pulmonary function for resectability.

Pathologic confirmation before systemic therapy. Multidisciplinary tumour board for borderline resectable IIIA/N2 disease — choose surgery-first vs definitive CRT vs neoadjuvant systemic.

- AJCC 9th nuances (updated T/N categories) — state that NCCN pathways now reference 9th edition.
- Never skip mediastinal staging in radiographic N2 before trimodality claims.
- Document smoking cessation — improves outcomes across modalities.

Early NSCLC staging & biomarker pathway

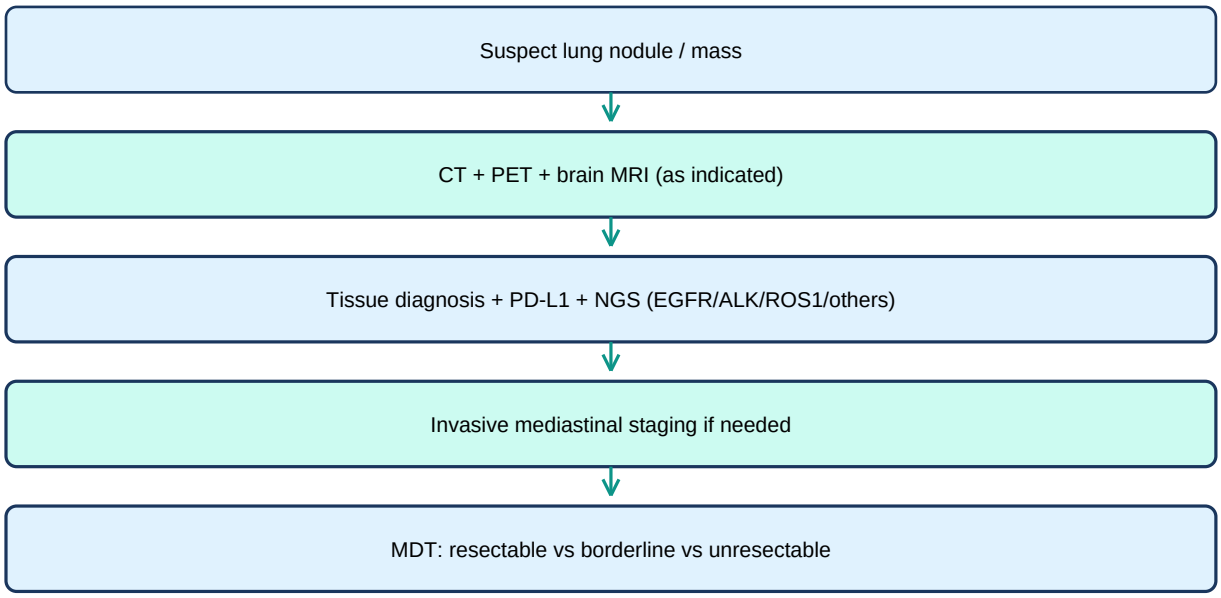


Figure: stepwise exam algorithm

3. Molecular testing in early-stage disease — why it changed practice

ADAURA and LAURA made EGFR genotyping mandatory for resected and unresectable stage III consideration of osimertinib. ALK and other drivers increasingly influence perioperative IO decisions (limited IO benefit / prefer targeted approaches when data mature). Broad NGS on resection specimen or pretreatment biopsy is best practice.

PD-L1 TPS still informs adjuvant IO discussions (IMpower010 enrichment) but perioperative trials enrolled broader populations — know each trial's biomarker subgroup.

EGFR ex19del / L858R	Adjuvant / post-CRT osimertinib sensitivity
PD-L1 / T-cell inflamed	IO benefit in perioperative & adjuvant settings
Driver+ caution	Prefer targeted path; IO benefit uncertain / lesser
MRD / ctDNA (emerging)	Risk stratification research — mention as future

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

• Write: 'Send EGFR before finalising adjuvant plan — ADAURA window.'

4. ADAURA — adjuvant osimertinib

ADAURA: stage IB–IIIA EGFR-mutated (ex19del/L858R) NSCLC after complete resection ± adjuvant chemo, randomised to osimertinib 80 mg daily vs placebo for 3 years. Dramatic DFS benefit (overall and stage II–IIIA); CNS DFS improved; OS benefit confirmed on longer follow-up — practice-changing adjuvant targeted therapy. ADAURA2 explores extended duration questions in selected residual-risk settings.

Practical: start after recovery from surgery/chemo; monitor labs/ECG as indicated; manage rash/diarrhoea/ILD rare risks; counsel adherence for 3 years.

Trial	Key result / exam takeaway

ADAURA	Adjuvant osi vs placebo EGFR+ IB–IIIA: DFS HR ~0.20–0.27 range in early reports; OS benefit later confirmed
ADAURA CNS	Marked reduction in CNS recurrence — key exam line
ADAURA2	Extended adjuvant osi questions in higher-risk residual disease

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Osimertinib adjuvant	80 mg PO daily × 3 years (ADAURA)

Table: doses and regimens

EXAM PEARLS / HALLMARKS

- OS confirmation elevated ADAURA from DFS-only criticism to firm standard.

5. PACIFIC — unresectable stage III consolidation durvalumab

PACIFIC: unresectable stage III NSCLC without progression after concurrent platinum CRT randomised to durvalumab vs placebo consolidation (up to 12 months). PFS and OS benefits established new standard for driver-negative (and historically all-comers) unresectable III. PD-L1 subgroup controversies exist for European labelling — discuss nuance: benefit clearest if PD-L1 ≥1% in some analyses, but practice varies by region.

Start ideally within 42 days of CRT; exclude severe unresolved pneumonitis. Concurrent IO+CRT regimens are evolving but PACIFIC consolidation remains the classic standard.

Trial	Key result / exam takeaway
PACIFIC	Durvalumab after cCRT: mPFS 16.8 vs 5.6 mo; 5-y OS ~43% vs 33%
PACIFIC-2 / others	Concurrent IO+CRT strategies — evolving, not yet universal replacement

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Durvalumab consolidation	10 mg/kg Q2W or 1500 mg Q4W × 12 months post-CRT (per label/practice)
cCRT backbone	Platinum doublet + 60 Gy-equivalent thoracic RT — centre protocols

Table: doses and regimens

6. LAURA — osimertinib after CRT in unresectable EGFR+ stage III

LAURA addressed EGFR-mutated unresectable stage III disease: after definitive CRT, osimertinib vs placebo markedly improved PFS — establishing targeted consolidation/maintenance rather than default durvalumab for EGFR+ patients. This pairs with ADAURA to make EGFR testing essential across early/locally advanced NSCLC.

Exam contrast table: PACIFIC (IO, all-comers/PD-L1) vs LAURA (EGFR TKI). Do not reflexively give durvalumab to EGFR+ unresectable III without considering LAURA.

Post-CRT consolidation fork

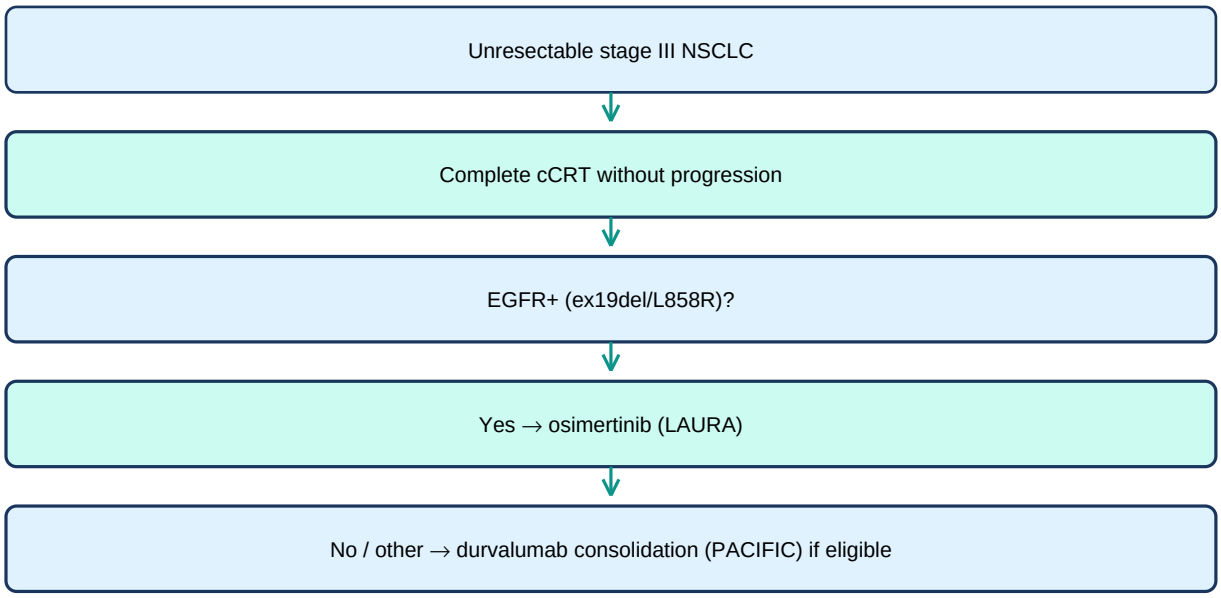


Figure: stepwise exam algorithm

Trial	Key result / exam takeaway
LAURA	Osi vs placebo after CRT EGFR+ unresectable III — major PFS win; new standard discussion
PACIFIC vs LAURA	Driver– → durvalumab; EGFR+ → osimertinib pathway

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Osimertinib (LAURA)	80 mg PO daily after CRT until progression/toxicity per protocol

Table: doses and regimens

7. CheckMate-816 — neoadjuvant chemo-IO

CheckMate-816: resectable stage IB–IIIA NSCLC randomised to neoadjuvant nivolumab + platinum chemo vs chemo alone. Improved pCR and EFS; surgery feasibility maintained. Established neoadjuvant chemo-IO as a preferred option for eligible driver-negative resectable disease, especially nodal-positive.

Practical pearls: restage after neoadjuvant; operate in window; path reports must include residual viable tumour percentage / pCR; manage irAEs without missing surgical timing.

Trial	Key result / exam takeaway
CheckMate-816	Neoadjuvant nivo+chemo vs chemo: higher pCR; EFS benefit — FDA neoadjuvant approval basis

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Nivolumab neoadjuvant	360 mg Q3W with platinum doublet × 3 cycles (816 schedule)
Platinum doublet	Histology-based (pemetrexed nonsquamous; paclitaxel/gemcitabine squamous variants)

Table: doses and regimens

8. KEYNOTE-671 & AEGEAN — perioperative IO platforms

KEYNOTE-671: perioperative pembrolizumab (neoadjuvant pembro+chemo → surgery → adjuvant pembro) vs perioperative chemo+placebo in resectable stage II–IIIB (N2). EFS and OS benefits — strong perioperative standard. AEGEAN: perioperative durvalumab + chemo improved EFS in resectable NSCLC. Together with 816, they redefine multimodality care.

Compare designs on exam: 816 = neoadjuvant only IO; 671/AEGEAN = peri-operative IO (before and after surgery). Choice may depend on approval, PD-L1, stage, and centre protocol.

Trial	Key result / exam takeaway
KEYNOTE-671	Perioperative pembro+chemo: EFS & OS benefit in resectable II–III
AEGEAN	Perioperative durvalumab+chemo: EFS benefit
CheckMate-77T	Perioperative nivo — additional perioperative evidence (mention if space)
Neotorch / other regional	Toripalimab perioperative data — global landscape awareness

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Pembrolizumab perioperative	200 mg Q3W with chemo neoadjuvant → adjuvant pembro per 671
Durvalumab perioperative	Per AEGEAN with platinum chemo → adjuvant durvalumab

Table: doses and regimens

EXAM PEARLS / HALLMARKS

- Say 'OS benefit in KEYNOTE-671' — differentiates from EFS-only trials.

9. Adjuvant immunotherapy after resection (IMpower010, KEYNOTE-091)

IMpower010: adjuvant atezolizumab after cisplatin-based chemo improved DFS in PD-L1 TC ≥1% stage II–IIIA (and stronger in TC ≥50%). KEYNOTE-091 / PEARLS: adjuvant pembrolizumab improved DFS in stage IB–IIIA after resection (± chemo). These remain relevant when patients did not receive neoadjuvant IO.

If perioperative IO already given, do not stack redundant adjuvant IO outside protocol. EGFR+ resected patients: prefer ADAURA osimertinib over adjuvant IO.

Post-operative systemic therapy fork

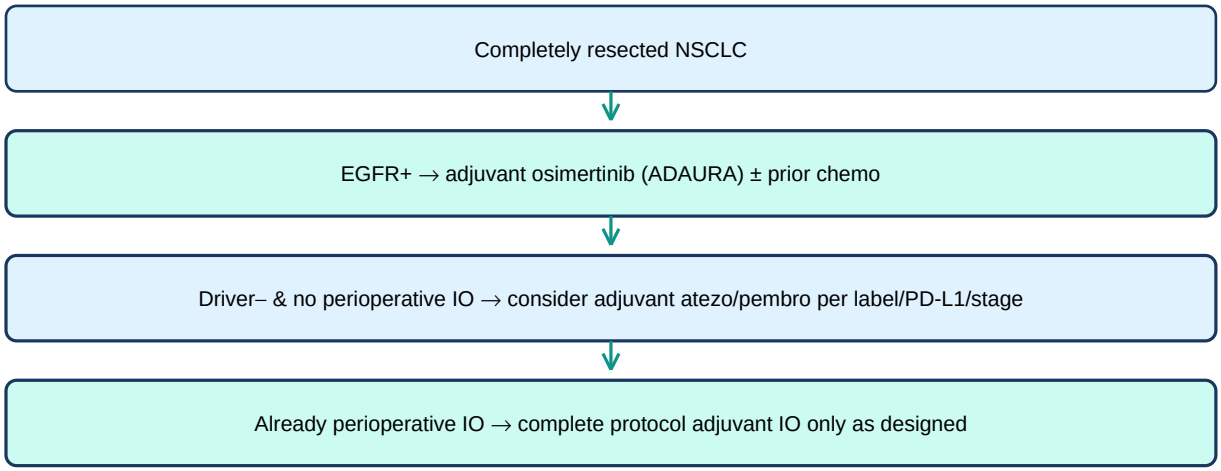


Figure: stepwise exam algorithm

Trial	Key result / exam takeaway
IMpower010	Adjuvant atezo after chemo — DFS benefit PD-L1+ II–IIIA
KEYNOTE-091 (PEARLS)	Adjuvant pembro — DFS benefit IB–IIIA
ADAURA vs adjuvant IO	EGFR+ → osi; driver– → IO pathway

Table: landmark clinical trials

10. Integrated management by stage (exam flowchart)

Synthesise into a single algorithm examiners can follow. Emphasise MDT, biomarkers first, then modality sequencing.

• Borderline N2: neoadjuvant systemic may convert / select biology (pCR).

• Incomplete resection: re-resection / RT / systemic per MDT — not covered by adjuvant trial populations cleanly.

• Pneumonitis risk: prior IO + thoracic RT needs careful PACIFIC timing.

Practice-changing early NSCLC master algorithm

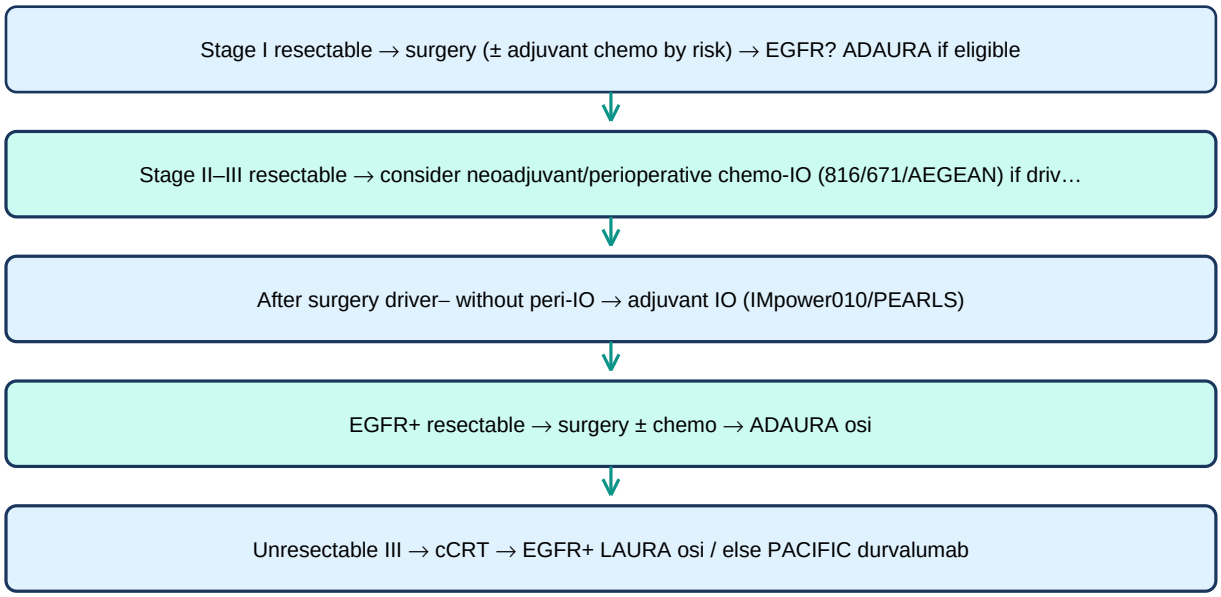


Figure: stepwise exam algorithm

11. Comparison table of practice-changing trials

Use this table as the centrepiece of the answer book.

Trial	Key result / exam takeaway
ADAURA	Adjuvant osi EGFR+ IB–IIIA — DFS + OS; 80 mg × 3 y
PACIFIC	Durvalumab post-cCRT unresectable III — PFS + OS
LAURA	Osi post-CRT EGFR+ unresectable III — PFS practice change
CheckMate-816	Neoadjuvant nivo+chemo — pCR + EFS
KEYNOTE-671	Perioperative pembro — EFS + OS
AEGEAN	Perioperative durvalumab — EFS
IMpower010	Adjuvant atezo after chemo — DFS in PD-L1+
KEYNOTE-091	Adjuvant pembro — DFS

Table: landmark clinical trials

12. Doses & regimens quick sheet

Examiners award marks for writable doses even in a 'advances' essay.

Drug / regimen	Dose / schedule (exam pearls)
Osimertinib	80 mg PO daily (ADAURA 3 years; LAURA post-CRT)
Durvalumab PACIFIC	10 mg/kg Q2W or 1500 mg Q4W × 12 months
Nivolumab 816	360 mg Q3W × 3 with platinum chemo pre-op
Pembrolizumab 671	200 mg Q3W perioperative schedule with chemo
Adjuvant atezolizumab	1200 mg Q3W after cisplatin chemo (IMpower010)
Adjuvant pembrolizumab	200 mg Q3W or 400 mg Q6W (PEARLS/label)

Table: doses and regimens

13. NCCN NSCLC v9.2026 updates (early & related)

Even though many 2026 bullets emphasise metastatic EGFR combinations, cite AJCC 9th adoption and reinforce biomarker-directed early pathways. Mention metastatic updates briefly only as 'continuum of EGFR care' if space (FLAURA2/MARIPOSA/Dato-DXd) without derailing the early-stage focus.

NCCN 2026 / 2027 UPDATES (include in answer)

• *Referenced versions:* Non-Small Cell Lung Cancer v9.2026

• AJCC 9th edition staging adopted in NCCN NSCLC pathway.

• Early/locally advanced care remains biomarker-stratified: EGFR testing to guide adjuvant osimertinib (ADAURA) and post-CRT osimertinib (LAURA) vs durvalumab (PACIFIC).

• Perioperative and neoadjuvant immunotherapy options incorporated for eligible resectable disease.

• Metastatic continuum (for context): osimertinib + platinum/pemetrexed and amivantamab + lazertinib Category 1 preferred 1L EGFR mut+; Dato-DXd preferred in selected post-osi+chemo settings — shows EGFR care continuity beyond early stage.

• ROS1 subsequent options (repotrectinib/taletrectinib) and HER2-mut ADC notes appear in advanced disease sections — test broadly on NGS.

EXAM PEARLS / HALLMARKS

• Lead NCCN paragraph with AJCC 9th — it is the explicit 2026 staging bullet.

14. Toxicities, practical pitfalls & Indian context

IO: irAEs may delay surgery — coordinate tightly. CRT+IO: pneumonitis differential (radiation vs immune). Osimertinib: rash, diarrhoea, rare ILD, QTc/lab monitoring. Access and cost of 3-year adjuvant osi and perioperative IO are real-world constraints — discuss NCG/institutional pathways and prioritisation of EGFR testing where TKI access exists.

- Do not give adjuvant IO instead of ADAURA in EGFR+.
- Do not skip surgery window after neoadjuvant IO due to unmanaged G1 toxicities.
- Document PD-L1 and EGFR on every pathology report for early NSCLC.

15. Summary — exam pearls wall chart

Close with a six-line wall chart mapping trial → setting → drug → endpoint.

- ADAURA: adjuvant osi EGFR+ — DFS/OS.
- PACIFIC: durvalumab post-CRT III — PFS/OS.
- LAURA: osi post-CRT EGFR+ III — PFS.
- 816 / 671 / AEGEAN: neoadjuvant & perioperative IO.
- IMpower010 / PEARLS: adjuvant IO if no peri-IO.
- AJCC 9th + universal molecular testing.

35-minute early NSCLC answer plan

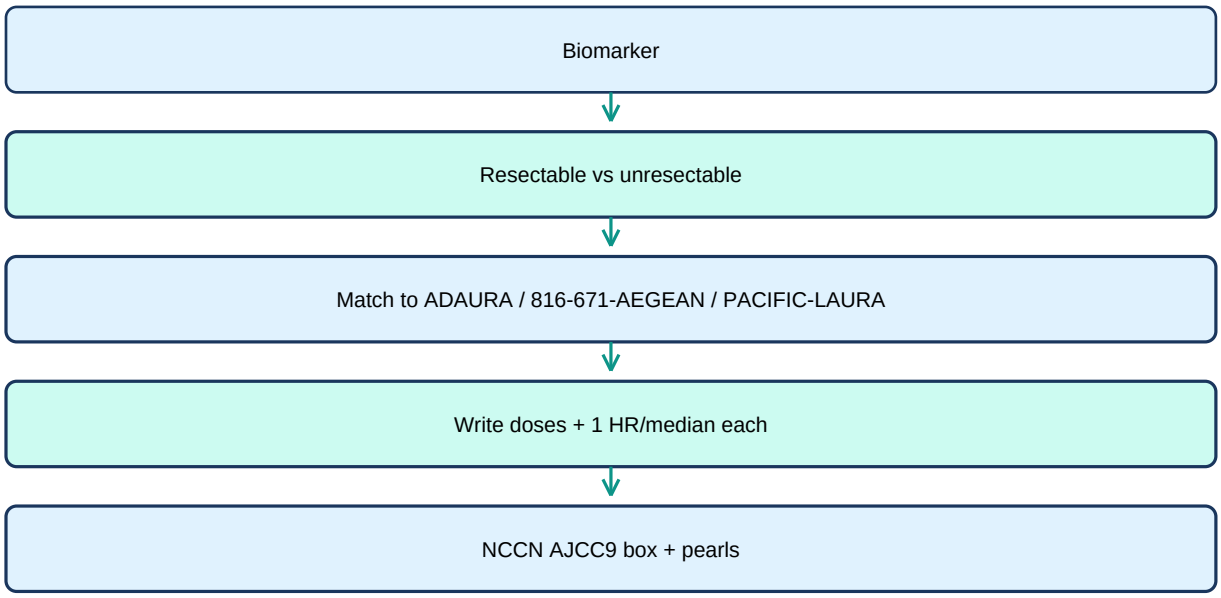


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- EGFR first — splits the entire algorithm.
- PACIFIC numbers: mPFS 16.8 vs 5.6 mo; quote 5-y OS if remembered.
- KEYNOTE-671 has OS — say so.
- LAURA prevents wrong default to durvalumab in EGFR+ III.
- AJCC 9th is the NCCN 2026 staging soundbite.

— Additional exam-length revision modules —

The following modules reinforce algorithms, toxicities, Indian/NCG context, and last-page summary structure.

3. Diagnosis, biomarkers & staging

A modern oncology answer must separate **diagnostic**, **prognostic**, and **predictive** markers. Predictive markers change drug choice (EGFR, HER2, BRCA, MSI, Claudin18.2, FGFR, NTRK).

- Tissue diagnosis with adequate IHC panel; never treat metastatic disease without molecular profiling when actionable targets exist.
- Use validated assays (FDA/companion diagnostics where mandated).
- Stage with AJCC 8th/9th as applicable (NSCLC NCCN 2026 uses AJCC 9th).
- Performance status, organ function, comorbidities, geriatric assessment, and goals of care before regimen selection.
- Broad NGS preferred: EGFR, ALK, ROS1, BRAF, MET ex14, RET, NTRK, KRAS, ERBB2, NRG1.
- PD-L1 TPS for immunotherapy decisions in driver-negative disease.

Work-up algorithm (draw this in the answer booklet)

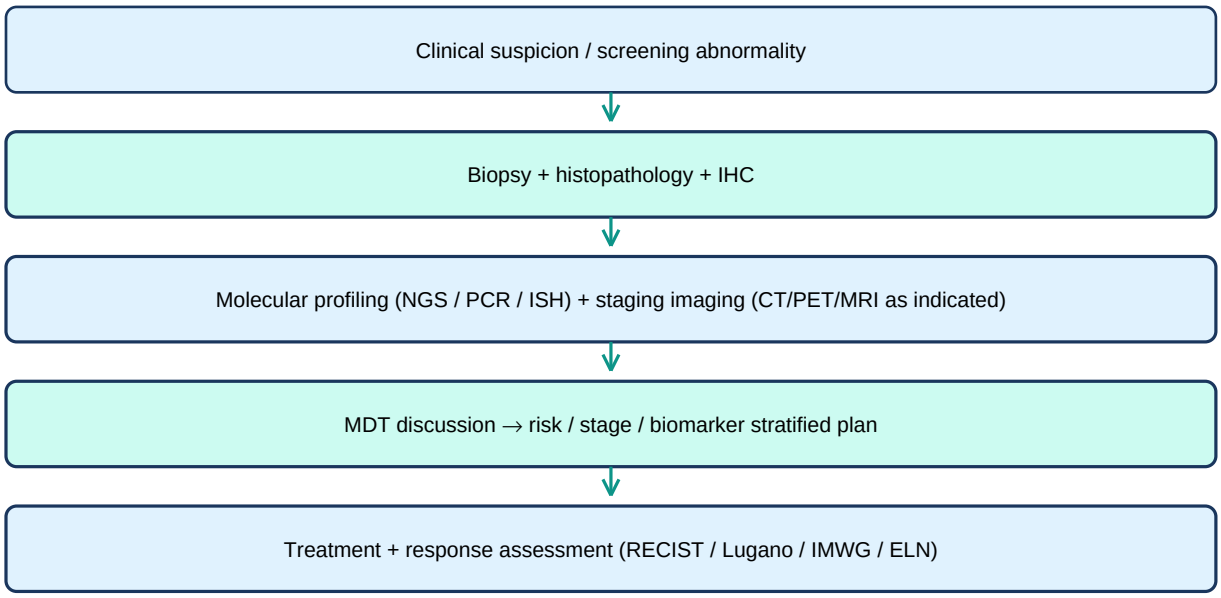


Figure: stepwise exam algorithm

4. Management algorithm, doses & regimens

Structure management as: (A) localized/curative-intent pathways, (B) advanced/metastatic systemic therapy, (C) special populations (CNS, elderly, organ dysfunction, pregnancy), (D) supportive care.

For each line of therapy state: eligibility, preferred regimen, dose, key toxicity monitoring, and the trial that supports it.

- Localized disease: multimodality — surgery/RT/systemic as per disease chapter.
- Neoadjuvant vs adjuvant: cite pathologic response and adjuvant switch trials (e.g., KATHERINE, CheckMate 577, KEYNOTE-522).
- Metastatic: biomarker-first; avoid empiric chemo when a high-level targeted/IO option exists.
- Oligometastatic disease: consider metastasis-directed therapy where NCCN supports (e.g., prostate APMS 2026).
- Re-biopsy / liquid biopsy at progression to map resistance.

Management flowchart (reproduce in exam)

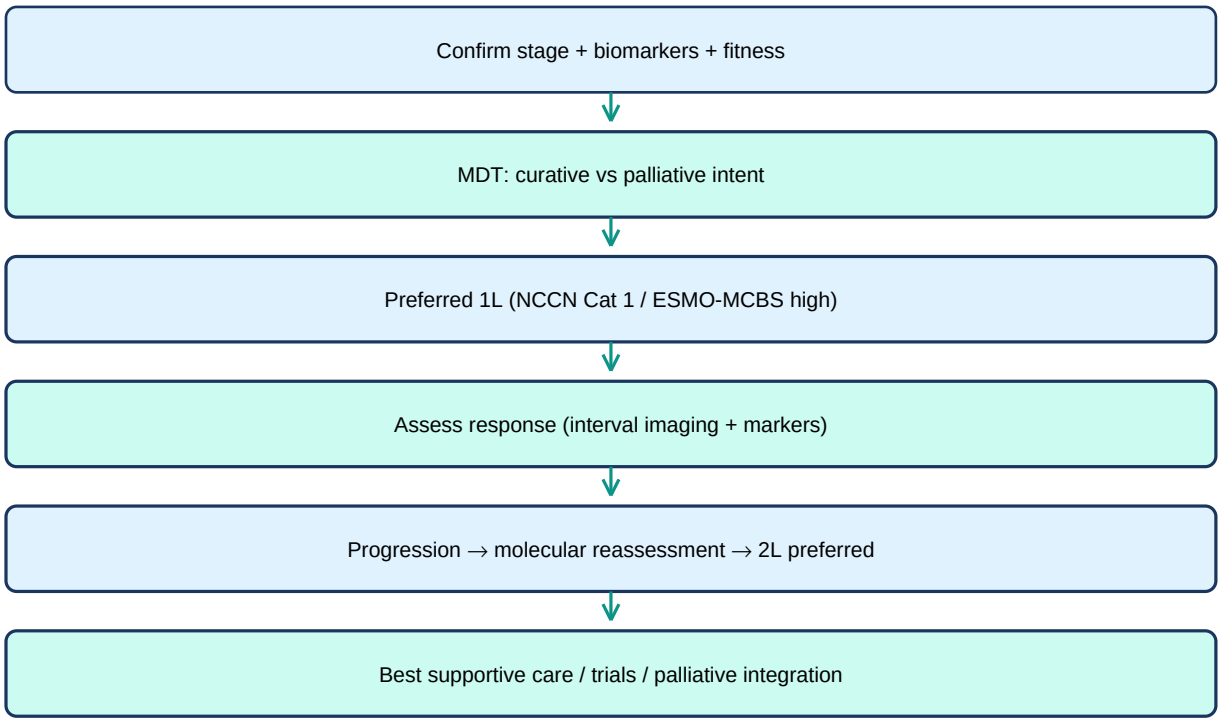


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)
Osimertinib	80 mg PO daily (adjuvant & metastatic); combine with platinum/pemetrexed per FLAURA2
Amivantamab	IV weight-based (typically 1050 mg <80 kg / 1400 mg ≥80 kg) weekly then Q2W; + lazertinib 240 mg PO daily (MARIPOSA)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Carboplatin AUC	AUC 5–6 with pemetrexed (NSCLC); AUC 5–7.5 ovarian regimens

Table: doses and regimens

5. Landmark clinical trials (must quote in answer)

Present trials in a table: name → design/population → intervention → primary endpoint → absolute/relative benefit → practice change. Examiners award marks for correct trial names and one numeric result (HR or median PFS/OS).

- Mention phase (III preferred for standards) and whether endpoint was PFS, OS, MRD, pCR, or EFS.
- Distinguish statistical vs clinical significance; note crossover and subgroup caveats.
- For India-focused answers, mention access/biosimilars/NCG guidance after the trial table.

Trial	Key result / exam takeaway
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MARIPOSA	Amivantamab + lazertinib vs osimertinib — PFS; NCCN Cat 1 preferred
ADAURA / ADAURA2	Adjuvant osimertinib stage IB–IIIA EGFR+
LAURA	Osimertinib after CRT in unresectable stage III EGFR+
CHRYSALIS / PAPILLON	Amivantamab (± chemo) for EGFR ex20ins
TROPION-Lung datasets	Dato-DXd activity → NCCN preferred post-osimertinib+chemo

Table: landmark clinical trials

6. Toxicities, supportive care & monitoring

A complete answer links efficacy to safety. List monitoring schedules (CBC, chemistry, ECG, TFTs, cortisol, LFTs, PFTs/CT for ILD risk drugs).

- Anticipate class toxicities and counsel patients; use CTCAE grading.
- Cardio-oncology: anthracyclines, HER2 mAbs, ibrutinib, checkpoint myocarditis — baseline ECG/echo when indicated.
- IO toxicities: irAEs — hold drug, steroids, specialty referral; endocrine irAEs often permanent replacement.
- ADC pulmonary toxicity (T-DXd ILD): hold permanently for ≥grade 2 in many protocols; imaging + steroids.
- TLS prophylaxis for venetoclax/BLIN/CAR-T; rasburicase if high-risk.
- Fertility, sexuality, survivorship, and psychosocial support are part of comprehensive answers.

Toxicity management loop

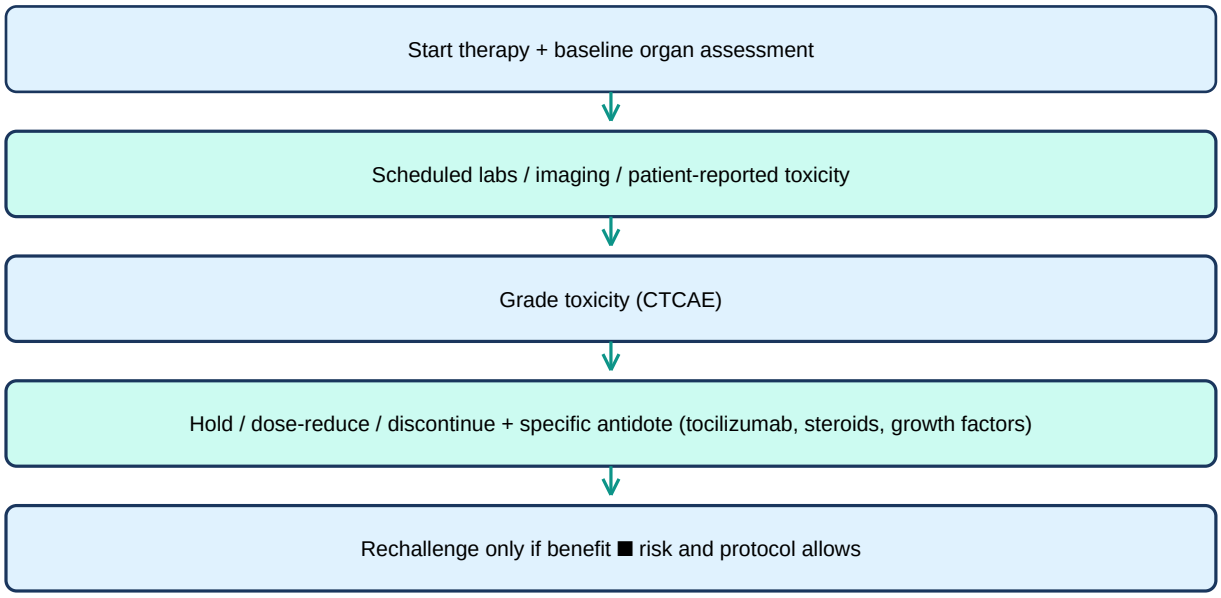


Figure: stepwise exam algorithm

7. NCCN 2026 / 2027 updates (high-yield for ‘recent advances’)

Paper IV rewards currency. Explicitly write: **“As per NCCN Guidelines updates in 2026/2027…”** and list 3–5 bullets. Also mention ESMO guidelines / Indian NCG where relevant for balance.

- State Category (1 / 2A / 2B) and preferred vs other recommended vs useful in certain circumstances.
- If Indian exam: reconcile NCCN with CDSCO approvals and NCG practical pathways.

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Non-Small Cell Lung Cancer v9.2026; Small Cell Lung Cancer v1.2027 / updates from v2.2026
- AJCC 9th edition staging adopted in NCCN NSCLC pathway.
- Osimertinib + platinum/pemetrexed AND amivantamab + lazertinib elevated to Category 1 preferred 1L for EGFR mut+ (detected before 1L systemic therapy).
- Datopotamab deruxtecan (Dato-DXd) preferred 2L after progression on osimertinib + chemotherapy; also preferred 3L after other preferred 1L regimens; listed for EGFR ex19del/L858R, uncommon EGFR, and EGFR ex20ins subsequent therapy.
- Amivantamab + lazertinib useful in certain circumstances after osimertinib progression (multiple lesions) if not previously given.
- Repotrectinib or taletrectinib preferred subsequent for ROS1+ with brain metastases.
- Ado-trastuzumab emtansine useful in certain circumstances for ERBB2/HER2-mutated NSCLC after frontline progression.
- Carboplatin + etoposide + atezolizumab → maintenance lurbinectedin + atezolizumab added for extensive-stage SCLC.
- Consider adding lurbinectedin to maintenance atezolizumab if ≥SD after 4 cycles induction, ECOG 0–1, no prior brain metastases.

• Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

EXAM PEARLS / HALLMARKS

• NCCN versions change frequently — cite year/version (e.g., NSCLC v9.2026, Breast v9.2026, Myeloma v2.2027).

• Category 1 = high-level evidence + uniform NCCN consensus.

8. Special situations, differentials & practical issues

Stage III unresectable: concurrent CRT then consolidative durvalumab (PACIFIC) if no driver contraindication; EGFR+ unresectable III → osimertinib (LAURA). Perioperative IO (CheckMate 816, KEYNOTE-671, AEGEAN) changed early NSCLC.

- Never smoke ≠ no drivers; test all nonsquamous and consider squamous with clinical features.
- CNS staging MRI important before TKI selection; prefer CNS-active agents.

9. Evidence hierarchy, patient rights & benefits

Quote evidence using: Phase III RCT > meta-analysis of RCTs > prospective registries / RWE > retrospective > expert consensus. NCCN Categories and ESMO-MCBS scores help communicate strength of benefit.

Patient-centered care: informed consent for toxicities (especially CRS/ICANS, ILD, infertility), shared decision-making, financial toxicity counseling, and access to supportive/palliative care concurrent with active therapy.

- Benefits to highlight: OS, PFS, pCR, MRD negativity, organ preservation, QoL, reduced hospitalization.
- Rights: autonomy, confidentiality (EMR), right to second opinion, trial participation, refusal of care, end-of-life preferences.
- Equity: molecular testing access, biosimilars, NCG/Ayushman Bharat practical pathways for Indian exams.
- Pharmacovigilance: report ADRs; Phase IV / post-marketing commitments.

EXAM PEARLS / HALLMARKS

• Write one sentence on absolute benefit (e.g., ‘median OS improved from X to Y months’).)

• Mention number-needed-to-treat conceptually if space permits.

10. Summary checklist (write this last page in the booklet)

Close the 15-mark answer with a crisp numbered summary the examiner can tick.

- Definition + epidemiology one-liner
- Key biology / hallmark diagram
- Essential diagnostics & biomarkers
- Risk/stage stratification
- Preferred treatment algorithm with ≥2 doses
- ≥3 named trials with endpoints
- Major toxicities + monitoring
- NCCN 2026 update bullets (3–5)
- Special situations
- Five exam pearls

30-second revision flowchart

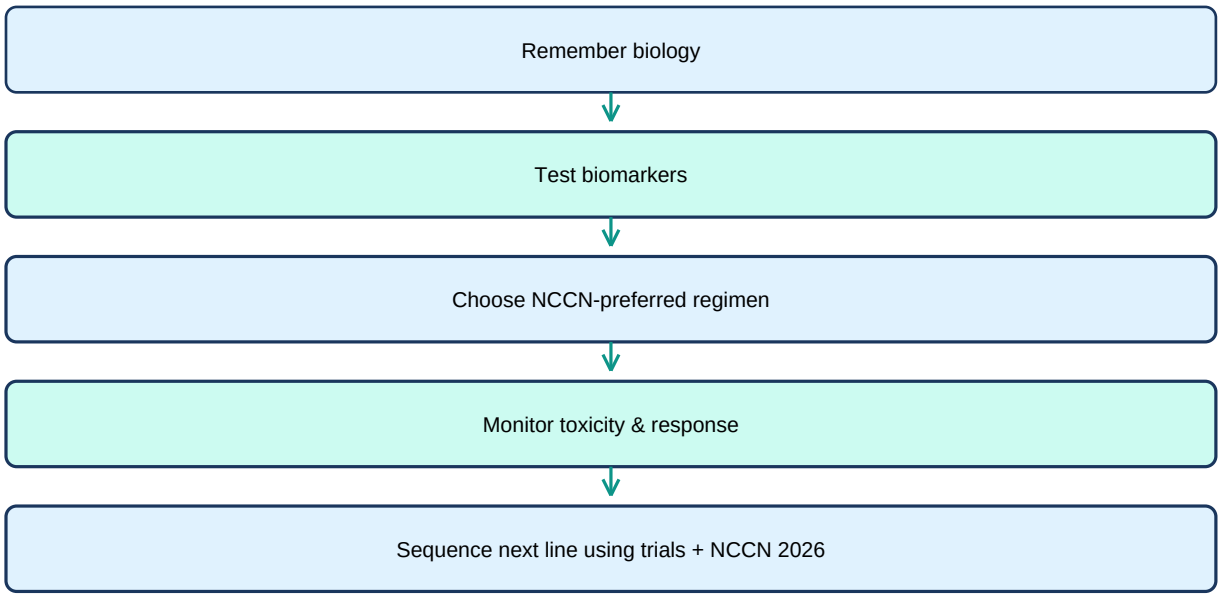


Figure: stepwise exam algorithm

EXAM PEARLS / HALLMARKS

- Neat flowchart + trial table often separates a 10/15 from a 13–15/15.
- Do not dump drug lists without doses or evidence.
- Currency: say ‘NCCN 2026’ explicitly for Paper IV.

11. Pharmacology deep-dive (class, PK/PD, interactions)

For drug-centered questions, present: chemical/biologic class → target epitope/kinase → PK (oral bioavailability, half-life, hepatic/renal clearance) → food effects → CYP interactions → dose adjustments → therapeutic drug monitoring if any.

Example interaction pearls: acid-suppressive therapy with some TKIs; QT-prolonging combinations; azole antifungals with vinca/TKIs; strong CYP3A inducers reducing exposure to many oral targeted agents.

- State renal/hepatic dose modifications when known.
- Note formulation differences (olaparib tablets vs capsules; IV vs SC daratumumab).
- Infusion reactions: premedication protocols for mAbs/BiTEs.

Drug / regimen	Dose / schedule (exam pearls)
Osimertinib	80 mg PO daily (adjuvant & metastatic); combine with platinum/pemetrexed per FLAURA2
Amivantamab	IV weight-based (typically 1050 mg <80 kg / 1400 mg ≥80 kg) weekly then Q2W; + lazertinib 240 mg PO daily (MARIPOSA)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Carboplatin AUC	AUC 5–6 with pemetrexed (NSCLC); AUC 5–7.5 ovarian regimens

Table: doses and regimens

12. Response assessment & follow-up schedules

Use disease-specific criteria: RECIST 1.1 solid tumors; iRECIST for immunotherapy; Lugano/Deauville for lymphoma; IMWG for myeloma; ELN molecular response for CML/AML; PSA working group for prostate.

- Typical imaging interval q2–3 cycles in metastatic disease unless clinically indicated sooner.

- PET-adapted strategies in Hodgkin lymphoma (RATHL, AHL2011) — cite Deauville 1–3 negative vs 4–5 positive.
- ctDNA MRD emerging for escalation/de-escalation (CRC, lymphoma, breast — exam discussion point).

Clinical algorithm

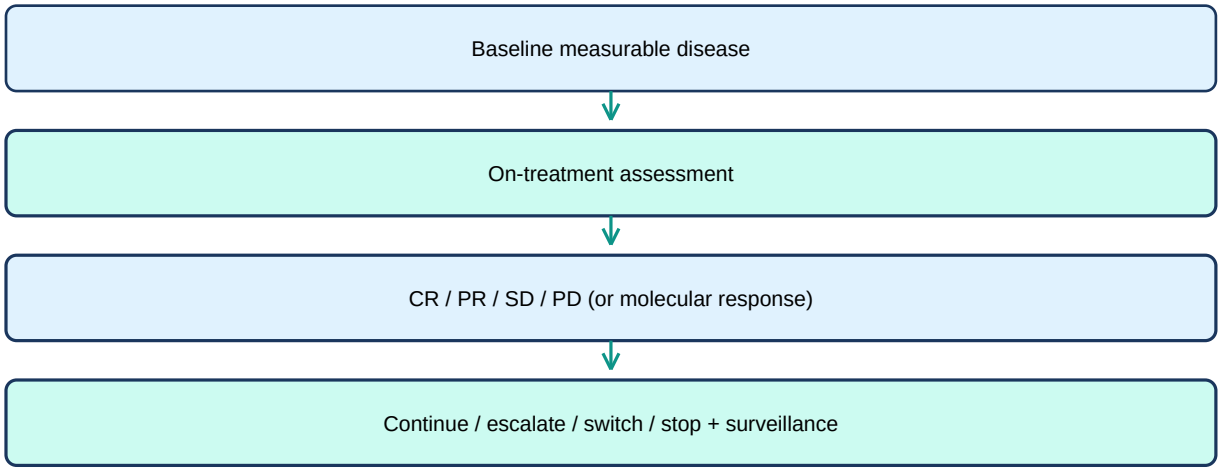


Figure: stepwise exam algorithm

13. Comparison tables useful for exams

Draw 2×N tables comparing old standard vs new standard; or drug A vs B (efficacy, toxicity, CNS activity, cost/access).

- Column ideas: target, dose, PFS, OS, grade ≥3 AEs, CNS activity, NCCN category.
- Another table: resistance mutation → preferred next drug.

Trial	Key result / exam takeaway
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MARIPOSA	Amivantamab + lazertinib vs osimertinib — PFS; NCCN Cat 1 preferred
ADAURA / ADAURA2	Adjuvant osimertinib stage IB–IIIA EGFR+
LAURA	Osimertinib after CRT in unresectable stage III EGFR+
CHRYSALIS / PAPILLON	Amivantamab (± chemo) for EGFR ex20ins

Table: landmark clinical trials

14. Indian context, NCG & practical exam tips

National Cancer Grid (NCG) resource-stratified guidelines are frequently expected in Indian DM exams. Discuss biosimilars (trastuzumab, rituximab), generic TKIs, and center-based cell therapy availability.

- Document MDT decisions; clinical trial referral when standard options exhausted.
- Counsel financial toxicity transparently; prioritize highest-value Category 1 options.
- Infection prophylaxis algorithms crucial in hematologic topics (antifungal, antiviral, PJP).

15. Sample ‘examiner hot spots’ & model closing paragraph

Model closing: “In summary, management of this problem is biomarker-driven. Current standards are supported by phase III trials and reflected in NCCN 2026/2027 updates. Sequencing should maximize OS and QoL while mitigating class-specific toxicities. Enrollment in clinical trials and adherence to MDT pathways remain integral.”

Hot spots: exact doses, one HR/median, resistance mechanism, NCCN category, and a drawn algorithm.

EXAM PEARLS / HALLMARKS

- If stuck on memory: write mechanism → class → one trial → one dose → one toxicity → NCCN 2026 line.
- Never contradict yourself between algorithm and trial table.

A. Epidemiology & burden (write ½–1 page)

Begin with global and Indian epidemiology when known (GLOBOCAN / ICMR / NCRP). State age distribution, sex ratio, and risk factors (tobacco, infection-related cancers such as HPV/HBV/HCV/H. pylori/HIV, obesity, inherited syndromes).

Explain why incidence or mortality trends matter for screening and resource allocation. For rare diseases, state incidence per 100,000 and referral-center patterns.

Link epidemiology to prevention: vaccination (HPV, HBV), tobacco control, hepatitis treatment, aspirin in selected molecular CRC, and lifestyle (exercise post-adjuvant CRC per NCCN 2026).

- One sentence on screening relevance (mammography, LDCT, FIT/colonoscopy, low-dose CT criteria).
- Mention hereditary fraction if applicable (Lynch, HBOC, FAP, MEN, VHL, HPRCC).
- State stage distribution at diagnosis in LMICs vs HIC if relevant to treatment sequencing.

B. Historical evolution → modern standard (timeline)

Examiners like a short timeline: historical surgery/RT/chemo era → targeted therapy era → immunotherapy/ADC/cellular therapy era.

Highlight inflection trials that changed guidelines (e.g., imatinib IRIS; trastuzumab HERA; checkpoint inhibitors; ADCs; CAR-T approvals).

End timeline with **NCCN 2026/2027** preferred regimens so the answer feels contemporary for Paper IV.

Therapeutic eras timeline

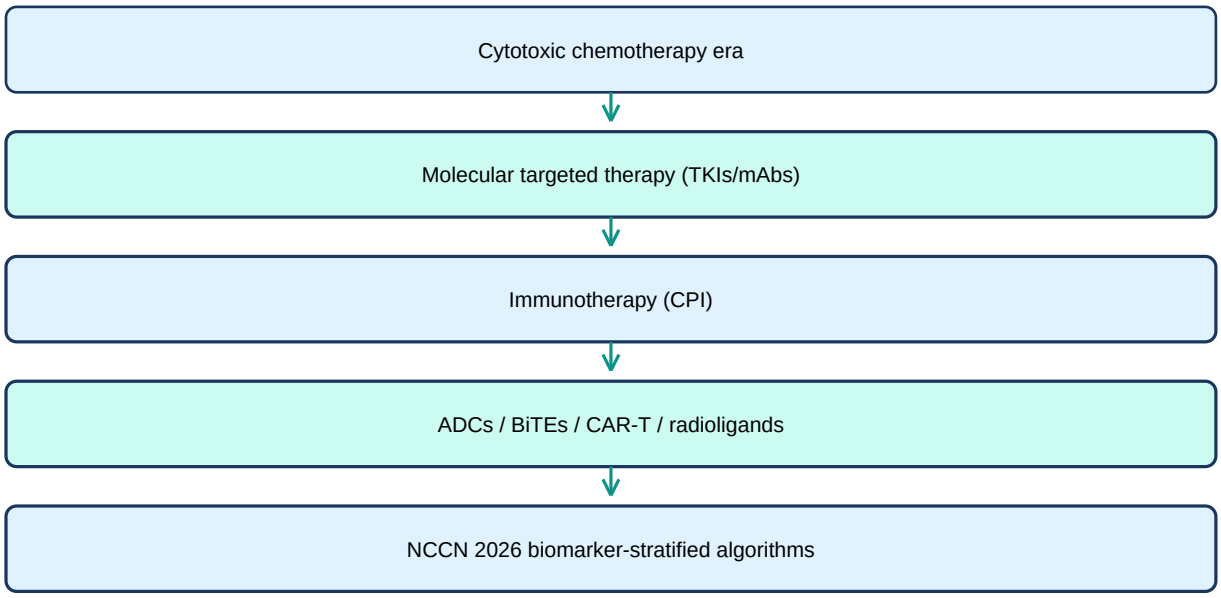


Figure: stepwise exam algorithm

C. Detailed differential diagnosis & pathology pearls

List 5–7 differentials and the IHC/molecular discriminators. Pathology reports should include grade, LVI, margins, nodes, and required biomarkers.

For hematologic topics: morphology → flow cytometry → cytogenetics → NGS panel → integrated WHO2022/ICC diagnosis.

For CUP/unknown primary style questions: IHC algorithm (CK7/CK20, TTF1, CDX2, GATA3, PAX8, synaptophysin) plus NGS and MSI/HER2.

- Always write ‘correlate clinically with radiology and tumor markers’.
- State pre-analytical caveats (decalcification, fixation time) affecting HER2/ER assays.
- Liquid biopsy complements but does not always replace tissue.

Morphology	Light microscopy pattern
IHC panel	Lineage + predictive markers
Cytogenetics	Karyotype / FISH fusions
NGS	SNVs, indels, CNAs, MSI, TMB
Integrated Dx	WHO/ICC + AJCC stage

Figure: mechanism / pathway map

D. Multidisciplinary care pathway & sequencing

Describe roles: medical oncology, surgical oncology, radiation oncology, radiology, pathology, palliative care, genetics, psycho-oncology, rehabilitation, and clinical trials unit.

Sequencing principles: treat drivers first when OS-proven; use neoadjuvant therapy to downstage and guide adjuvant escalation; avoid overlapping severe toxicities; protect future options (e.g., cumulative cardiotoxicity, neuropathy).

For oligometastatic disease discuss local therapy (surgery/SBRT/ablation) plus systemic therapy — prostate NCCN 2026 explicitly incorporates metastasis-directed therapy in selected APMS disease.

- Document MDT minutes in the file — also good exam phrase.
- Re-staging after neoadjuvant before definitive local therapy.
- Early palliative care improves QoL and sometimes survival in metastatic settings.

Clinical algorithm

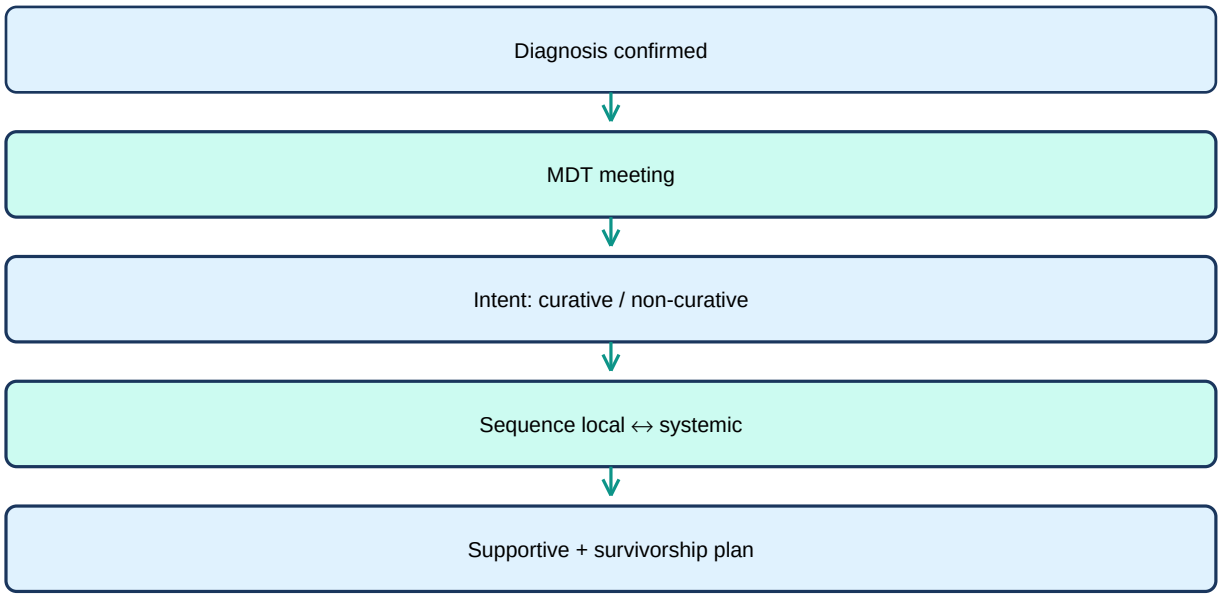


Figure: stepwise exam algorithm

E. Thoracic oncology deep notes

Molecular first in advanced nonsquamous NSCLC; consider in squamous with never-smoker/young features. PD-L1 TPS guides IO if driver-negative.

EGFR 1L options 2026: osimertinib alone; osimertinib+platinum/pemetrexed (FLAURA2); amivantamab+lazertinib (MARIPOSA) — NCCN Cat 1 preferred combinations.

Post-progression after osi+chemo: Dato-DXd preferred (NCCN 2026). Resistance testing: tissue/liquid for MET, C797X, fusions, histologic transformation.

Early NSCLC: adjuvant osimertinib (ADAURA); perioperative IO trials; unresectable III durvalumab (PACIFIC) vs osi (LAURA) if EGFR+.

SCLC ES: chemo-IO then consider lurbinectedin+atezolizumab maintenance per NCCN 2026 update in selected patients.

Clinical algorithm

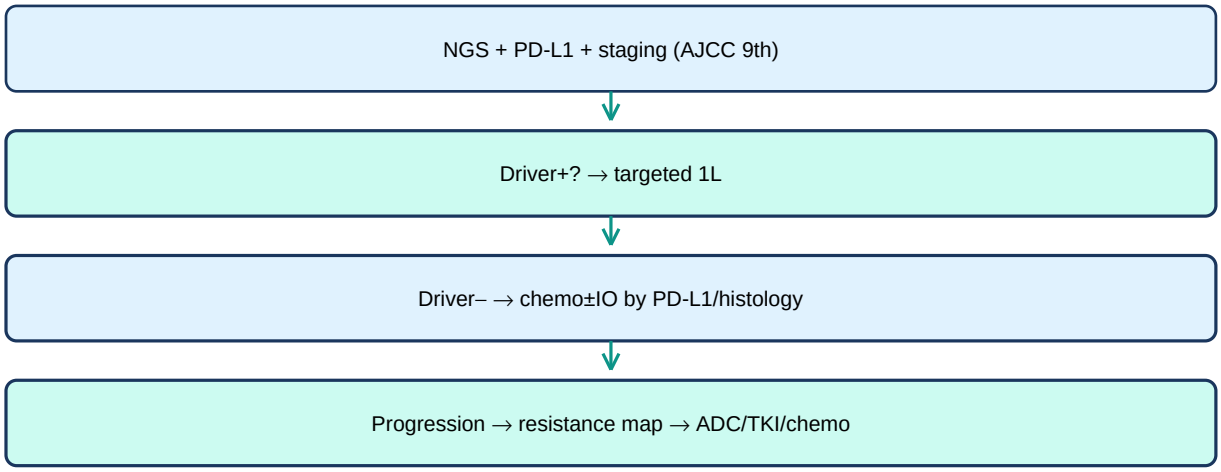


Figure: stepwise exam algorithm

Drug / regimen	Dose / schedule (exam pearls)

Osimertinib	80 mg PO daily
FLAURA2 chemo	Platinum + pemetrexed with osi
Amivantamab + lazertinib	IV amivantamab + lazertinib 240 mg PO daily
Dato-DXd	TROP2 ADC — preferred subsequent per NCCN 2026 EGFR pathway

Table: doses and regimens

G. Supportive care mini-chapter

Antiemetics (olanzapine-containing regimens for HEC), G-CSF primary/secondary prophylaxis, antimicrobial prophylaxis, mucositis care, nutrition (including TPN indications/contraindications), pain ladder + interventional techniques, thrombosis prevention (Khorana), growth factors, transfusion thresholds.

Psychosocial distress screening, fertility preservation referral before gonadotoxic therapy, and survivorship clinics.

- Febrile neutropenia: risk assess; anti-pseudomonal beta-lactam; add antifungal if persistent.
- Checkpoint diarrhea: rule out infection then steroids promptly.
- ADC ILD algorithms — patient education on dyspnea.

H. Model answer skeleton (time plan for 35 minutes)

Minutes 0–3: definition + epidemiology + diagram outline. Minutes 3–12: biology + diagnosis. Minutes 12–25: management algorithm with doses/trials. Minutes 25–30: toxicities + NCCN 2026. Minutes 30–35: summary pearls + underline Category 1 regimens.

- Leave a half page blank early for a neat flowchart drawn once.
- Underline drug doses.
- Box the NCCN 2026 bullets.

EXAM PEARLS / HALLMARKS

- Currency + clarity > exhaustive rare regimens.
- Named trial + number beats unnamed claim.
- NCCN 2026 phrase is expected in Paper IV recent advances.

K. Pitfalls, contraindications & ‘do not miss’ warnings

List absolute/relative contraindications of key drugs (e.g., baseline ILD caution for T-DXd; severe autoimmune disease caution for CPI; T315I needs ponatinib/asciminib not 1G/2G TKI alone; ATRA differentiation syndrome in APL; TLS with venetoclax/Blincyto; HBV reactivation with anti-CD20/CPI).

Drug–drug interactions and QT prolongation traps are classic viva points — mention checking concomitant meds (azoles, macrolides, acid suppressants with selected TKIs).

Imaging pitfalls: pseudo-progression on IO; flare on bone scan; PET Deauville interpretation after immunomodulatory therapy.

- Always screen hepatitis B before anti-CD20, anthracyclines, and many systemic regimens.
- Pregnancy test before teratogenic agents (IMiDs — REMS; MTX; TKI counseling).
- Do not combine overlapping QT-prolonging agents without ECG plan.
- Document performance status — unfit vs fit changes entire AML/lymphoma pathway.

Clinical algorithm

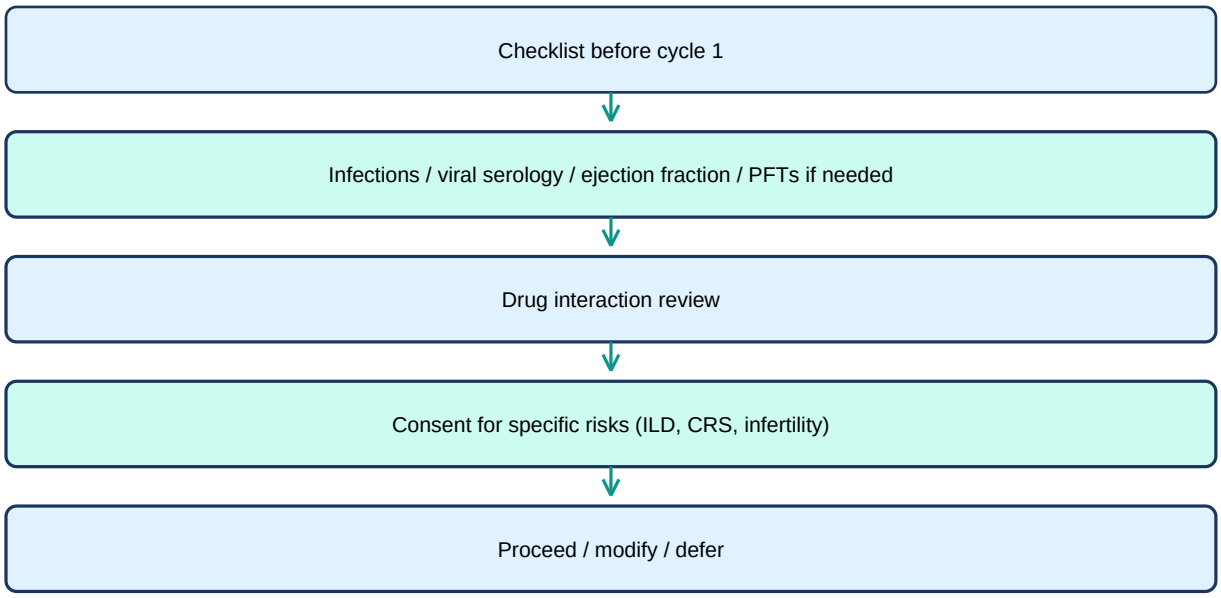


Figure: stepwise exam algorithm

L. Case vignette practice (apply the algorithm)

Invent a short case in the answer: age, stage, biomarkers, comorbidities → your exact regimen with dose → expected monitoring → what you will do at progression. This shows applied knowledge and often gains examiner goodwill.

Example template: “45-year-old, ECOG 1, biomarker X positive, no contraindication Y → start regimen Z at dose… per trial T / NCCN 2026 preferred. Assess response at interval N. At progression, re-profile for resistance mutation R and switch to agent S.”

Add one geriatric or organ-dysfunction variant to show nuance (dose reduce, alternate preferred regimen, or best supportive care).

- Vignette 1: fit, biomarker-positive — preferred Cat 1 therapy.
- Vignette 2: comorbid/unfit — alternative regimen.
- Vignette 3: progression with resistance mutation — sequenced next line.
- Vignette 4: severe toxicity — hold, treat, rechallenge decision.

M. One-page ‘wall chart’ summary (draw/box this)

Reserve the penultimate page for a wall-chart style summary with four quadrants: (1) Biology (2) Work-up (3) Treatment lines (4) NCCN 2026 bullets. This mirrors how many toppers present long answers and helps hit the page target with high signal density.

Quadrant 1	Hallmark gene / pathway / figure
Quadrant 2	Must-order tests + staging
Quadrant 3	1L / 2L / 3L with doses
Quadrant 4	Trials + NCCN 2026 + toxicities
Center	MDT + patient goals

Figure: mechanism / pathway map

EXAM PEARLS / HALLMARKS

- Wall chart + flowchart + trial table = visual completeness.
- Write 'NCCN 2026' in quadrant 4 explicitly.

I2. Full-length essay addendum — tables to copy into the answer book

For 15-mark elaborations, add two hand-drawn tables: (Table A) Drug–dose–pearl; (Table B) Trial–population–endpoint–practice change. Then write a short paragraph on quality-of-life and patient-reported outcomes, which modern trials increasingly report.

Discuss biomarker evolution on therapy (liquid biopsy cadence), and when to stop therapy (duration of CPI, TFR in CML, maintenance stopping rules).

Include a brief note on second malignancies and late effects when curative-intent therapy is discussed (alkylators, RT fields, PARPi MDS/AML risk, CHL survivors).

- Table A columns: Drug | Dose | Key toxicity | Pearl
- Table B columns: Trial | Arm | Primary endpoint | Result | Change in practice
- One paragraph on shared decision-making and financial counseling

Trial	Key result / exam takeaway
FLAURA	Osimertinib vs 1st-gen EGFR TKI — PFS/OS 1L
FLAURA2	Osimertinib + platinum/pemetrexed — PFS benefit; NCCN Cat 1 preferred
MARIPOSA	Amivantamab + lazertinib vs osimertinib — PFS; NCCN Cat 1 preferred
ADAURA / ADAURA2	Adjuvant osimertinib stage IB–IIIA EGFR+
LAURA	Osimertinib after CRT in unresectable stage III EGFR+
CHRYSALIS / PAPILLON	Amivantamab (± chemo) for EGFR ex20ins
TROPION-Lung datasets	Dato-DXd activity → NCCN preferred post-osimertinib+chemo

Table: landmark clinical trials

Drug / regimen	Dose / schedule (exam pearls)
Osimertinib	80 mg PO daily (adjuvant & metastatic); combine with platinum/pemetrexed per FLAURA2
Amivantamab	IV weight-based (typically 1050 mg <80 kg / 1400 mg ≥80 kg) weekly then Q2W; + lazertinib 240 mg PO daily (MARIPOSA)
Pembrolizumab	200 mg IV Q3W or 400 mg Q6W
Carboplatin AUC	AUC 5–6 with pemetrexed (NSCLC); AUC 5–7.5 ovarian regimens

Table: doses and regimens

NCCN 2026 / 2027 UPDATES (include in answer)

- *Referenced versions:* Non-Small Cell Lung Cancer v9.2026; Small Cell Lung Cancer v1.2027 / updates from v2.2026
- AJCC 9th edition staging adopted in NCCN NSCLC pathway.
- Osimertinib + platinum/pemetrexed AND amivantamab + lazertinib elevated to Category 1 preferred 1L for EGFR mut+ (detected before 1L systemic therapy).
- Datopotamab deruxtecan (Dato-DXd) preferred 2L after progression on osimertinib + chemotherapy; also preferred 3L after other preferred 1L regimens; listed for EGFR ex19del/L858R, uncommon EGFR, and EGFR ex20ins subsequent therapy.
- Amivantamab + lazertinib useful in certain circumstances after osimertinib progression (multiple lesions) if not previously given.
- Repotrectinib or taletrectinib preferred subsequent for ROS1+ with brain metastases.
- Ado-trastuzumab emtansine useful in certain circumstances for ERBB2/HER2-mutated NSCLC after frontline progression.
- Carboplatin + etoposide + atezolizumab → maintenance lurbinectedin + atezolizumab added for extensive-stage SCLC.
- Consider adding lurbinectedin to maintenance atezolizumab if ≥SD after 4 cycles induction, ECOG 0–1, no prior brain metastases.

• Expanding ADC footprint across NCCN 2026: T-DXd (HER2 breast/gastric/NSCLC/urothelial), Dato-DXd (EGFR+ NSCLC), sacituzumab govitecan (breast/urothelial), mirvetuximab (ovarian FRα), enfortumab vedotin (urothelial), belantamab combinations (myeloma), loncastuximab (B-cell).

I. Extended notes — resistance, sequencing & future directions

Resistance biology should occupy a dedicated half-page: on-target secondary mutations, bypass signaling, lineage plasticity (e.g., EGFR NSCLC → SCLC), antigen loss after cellular therapy, and immunosuppressive TME changes after CPI.

Sequencing principles: use the agent with largest OS benefit early when toxicities allow; reserve cross-resistant drugs; consider cumulative neuropathy/cardiotoxicity; plan CNS-active agents if brain metastases are present or likely.

Future directions appropriate for 2026 answers: tumor-agnostic indications (NTRK, RET, BRAF, MSI-H, TMB-H, HER2-mut), radioligand therapy, CAR-NK/off-the-shelf products, ctDNA-guided escalation/de-escalation, AI decision-support, and microbiome modulation adjuncts.

Health services angle: real-world evidence validating trial results in older/comorbid patients; digital remote toxicity monitoring; and equity of molecular testing access — themes also reflected in recent Paper IV questions.

- Write one ‘if progression then…’ resistance table.
- Mention at least one tumor-agnostic approval if biology overlaps.
- Close with trial enrollment as a standard option.

J. Quick-revision flash cards (last 1–2 pages)

Convert the topic into 8–10 flash cards mentally: (1) definition (2) hallmark gene (3) key test (4) 1L preferred (5) dose (6) trial (7) top toxicity (8) NCCN 2026 update (9) Indian/NCG pearl (10) one differential.

- Flash 1: One-line definition
- Flash 2: Driver / pathway
- Flash 3: Mandatory biomarker
- Flash 4: NCCN preferred 1L
- Flash 5: Exact dose
- Flash 6: Landmark phase III + endpoint
- Flash 7: Grade ≥3 toxicity to warn
- Flash 8: 2026 guideline change
- Flash 9: Resistance next step
- Flash 10: Supportive care must-do

EXAM PEARLS / HALLMARKS

- If time-poor in the hall: draw algorithm + trial table + NCCN box — captures most marks.
- Keep handwriting large for doses and HRs.

Final box — paste into last page of answer sheet

Re-list NCCN 2026 updates here so the examiner sees currency even if earlier pages are rushed.

NCCN 2026 / 2027 UPDATES (include in answer)

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Final NCCN 2026 box

Copy these bullets onto the last page of your answer booklet for Paper IV currency marks.

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Disclaimer: Compiled for DM exam revision from published trials and NCCN 2026/2027 public summaries (ASCO Post, OncLive, JNCCN selections, NCCN category list). Always verify the current full NCCN PDF, FDA/CDSCO labels, and institutional protocols before patient care. Doses require BSA/organ-function adjustment.